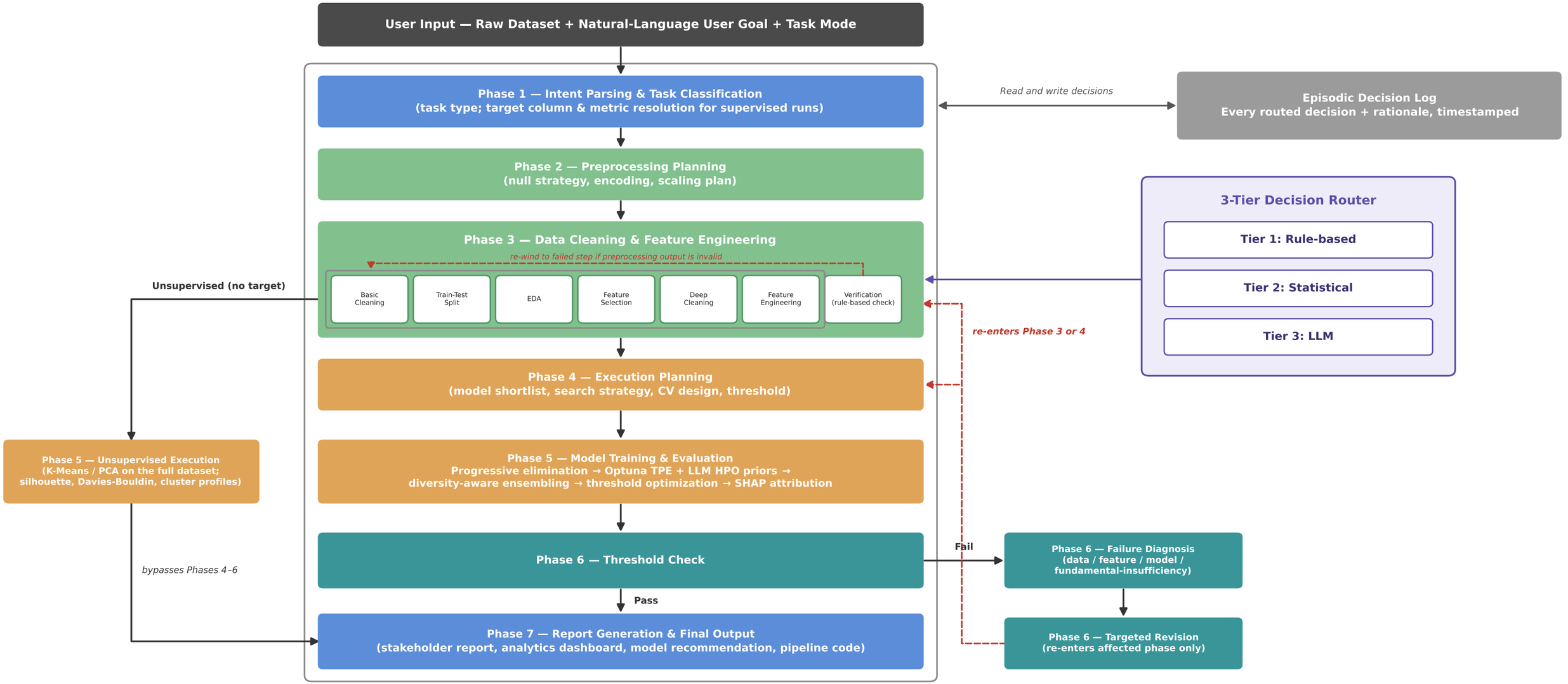
User Input — Raw Dataset + Natural-Language User Goal + Task Mode
Phase 1 — Intent Parsing & Task Classification
(task type; target column & metric resolution for supervised runs)
Phase 2 — Preprocessing Planning
(null strategy, encoding, scaling plan)
Phase 3 — Data Cleaning & Feature Engineering
re-wind to failed step if preprocessing output is invalid
Basic Cleaning
Train-Test Split
EDA
Feature Selection
Deep Cleaning
Feature Engineering
Verification (rule-based check)
Phase 4 — Execution Planning
(model shortlist, search strategy, CV design, threshold)
Phase 5 — Model Training & Evaluation
Progressive elimination → Optuna TPE + LLM HPO priors →
diversity-aware ensembling → threshold optimization → SHAP attribution
Phase 6 — Threshold Check
Pass
Phase 7 — Report Generation & Final Output
(stakeholder report, analytics dashboard, model recommendation, pipeline code)
Read and write decisions
Episodic Decision Log
Every routed decision + rationale, timestamped
3-Tier Decision Router
Tier 1: Rule-based
Tier 2: Statistical
Tier 3: LLM
re-enters Phase 3 or 4
Unsupervised (no target)
Phase 5 — Unsupervised Execution
(K-Means / PCA on the full dataset;
silhouette, Davies-Bouldin, cluster profiles)
bypasses Phases 4–6
Fail
Phase 6 — Failure Diagnosis
(data / feature / model /
fundamental-insufficiency)
Phase 6 — Targeted Revision
(re-enters affected phase only)